Y3S2 Group Project Explaining the Code

Jack D. Maher (jxm1517)

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1 Pre-processing

1.1 Atmospheric Assumptions and Composition

1.1.1 One-Dimensional Isothermal Atmosphere

When forming a model absorption spectrum, our script takes the input planetary parameters and assumes a one-dimensional, isothermal atmosphere. The one-dimensional approximation neglects latitudinal and longitudinal variations, treating the atmosphere as spherically symmetric. This means we do not see the effects of day-night temperature contrasts or banding that can exist in real exoplanet atmospheres.

The assumption of a one-dimensional atmosphere is common in exoplanet atmospheric modeling [1, 2]. and is a reasonable approximation when the goal is to understand general trends in the spectrum.

1.1.2 Ideal Gas Assumption

The atmosphere is modeled as an ideal gas. Under this assumption, the number density (n) is calculated from the ideal gas law, equation 1, which relates pressure (P), temperature (T), and number density (n) where k_B is the Boltzmann constant.

$$n = \frac{P}{k_B T} \tag{1}$$

This assumption is reasonable for exoplanet atmospheres at the pressures and temperatures typically probed in transmission spectroscopy. The ideal gas law allows us to directly relate pressure measurements to particle number densities, which are essential for calculating opacity and optical depth.

Under assumptions 1.1.1 and 1.1.2 , the pressure (P) decreases with radial distance (r) according to equation 2, reflecting how pressure falls off in a hydrostatic atmosphere moving outward from the planet.

$$P(r) \propto \frac{1}{r^2} \tag{2}$$

1.1.3 Constant Composition

The atmosphere is also assumed to have a uniform mixing ratio for each molecules in the atmosphere meaning the atmosphere's composition does not vary with altitude or pressure, although the density of the gasses will decrease with altitude as discussed in Section 1.1.2. In reality, atmospheric composition can vary with height due to diffusive separation and chemical processes [3], but this approximation is reasonable and is a standard assumption used in other exoplanet atmospheric studies [4].

Having a constant composition simplifies the calculation significantly, allowing us to specify the atmosphere's composition once and apply it uniformly across all pressure levels.

1.1.4 Cloud Deck

The cloud deck (at a specified pressure P_{cloud}) is treated as a hard cutoff and is assumed to be 100% opaque below that level. This represents an idealized cloud deck where all light is completely blocked below a sharp pressure boundary. This assumption effectively sets an inner radius r_{cloud} below which no light is transmitted.

This is a reasonable approximation for clouds in exoplanet atmospheres, which can often be optically thick and produce a sharp cutoff in the transmission spectrum.

1.2 Radiative Transfer and Optical Depth

When the planet passes in front of the star, a fraction of light rays pass through the upper atmosphere so are weakly attenuated, however, deeper rays are blocked by the opaque interior either clouds or solid surface and cut off the transmission spectrum at that depth. A visual diagram of this process is shown in Figure 1 where the star's light is attenuated as it passes through the planet's atmosphere, with different layers contributing to the total optical depth based on their density.

The aim of this function in our script is to calculate how much starlight is removed along each ray at each wavelength.

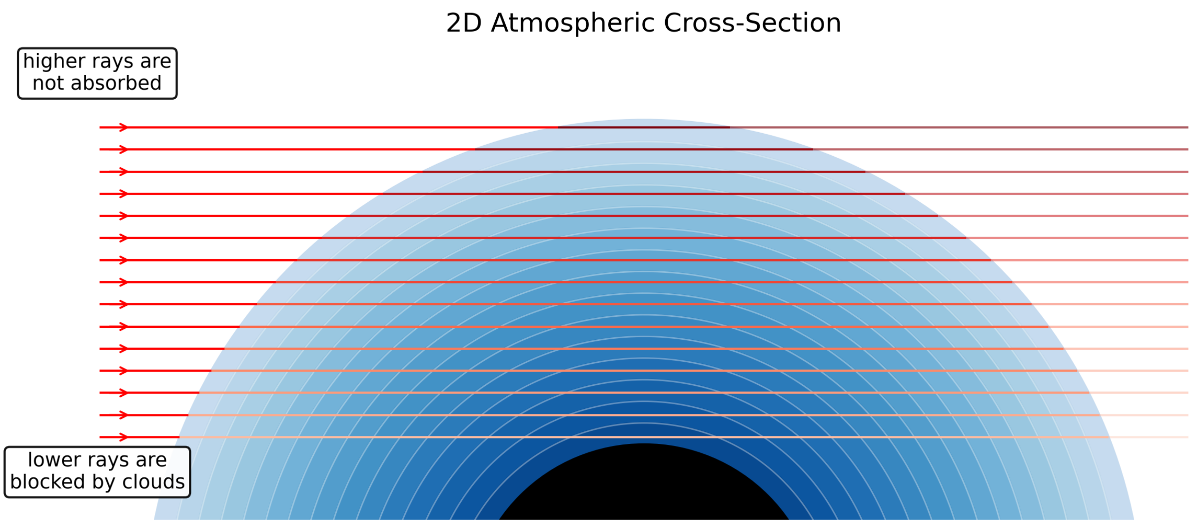

Figure 1: Diagram illustrating the radiative transfer process in an exoplanet atmosphere.

We calculate attenuation of the incoming ray, which is written in equation 3 as a function of the optical depth τ_λ along the ray, where $I_{\lambda,0}$ is the incoming specific intensity from the star, s is the distance along the ray, and τ_λ is the optical depth.

$$I_\lambda(s) = I_{\lambda,0} e^{-\tau_\lambda(s)} \tag{3}$$

If $\tau_\lambda \ll 1$ the atmosphere is mostly transparent at that wavelength; if $\tau_\lambda \gg 1$ the ray is mostly blocked.

The script then builds the wavelength-dependent optical depth from small path elements using the relation in equation 4 where κ_λ is the mass absorption coefficient, ρ is the density, n_i is the number density of species

i, and $\sigma_{\lambda,i}$ is its absorption cross section.

$$d\tau_\lambda = \kappa_\lambda \, \rho \, ds = \sum_i n_i \, \sigma_{\lambda,i} \, ds \tag{4}$$

Each pressure layer contributes an amount of absorption set by how many particles are present and how strongly they absorb at that wavelength. Summing these contributions along the full ray of light, which is done at the end of the pre-processing phase, gives τ_λ for that impact parameter.

1.3 Pressure Grid and Hydrostatic Equilibrium

The code represents the exoplanet atmosphere as a 1D logarithmic pressure grid from P = 10^{-4}Pa to 10^7Pa. This range covers the pressures probed probed in transmission spectroscopy, from the thin upper atmosphere to the deeper layers where clouds or surfaces may exist. While it could be argued that the limit for high pressure is too great, as it is unrealistic that we would be able to probe such depths, we believe it is necessary to ensure that the cloud deck is included in the grid for cases where P_{cloud} is high. For these cases however, the large amount For each pressure level P_j it assigns a temperature (T_j) using equation 5, where T_p is a reference temperature and P_j is the pressure at that level, and number density using the ideal gas law equation 1.

$$T_j = T_p P_j \tag{5}$$

To know how far rays travel through each layer, we need to find the radius r at each pressure level. This is obtained by integrating the equation of hydrostatic equilibrium, $\frac{\mathrm{d}P}{\mathrm{d}r}$, which leads to the expression in equation 6.

$$dr = -\frac{k_B T}{\mu g} dlnP \tag{6}$$

The calculation of gravity relative to radius, where g_{ref} is the surface gravity at a reference pressure P_{ref} and $r_{\text{ref}} = R_{\text{p}}$ is the corresponding radius, is stated in equation 7,

$$g(r) = g_{\text{ref}} \left(\frac{r_{\text{ref}}}{r}\right)^2 \tag{7}$$

From this reference point, the script steps outwards and inwards in even steps of $\log P$ to build the arrays $r(P_j)$ and $g(P_j)$. This converts the original pressure grid into a grid of atmospheric shells.

We then set the composition of the atmosphere in terms of logarithmic mixing ratios ($\log_{10} X_i$) for each molecule. This was calculated as a typical sub-neptune's composition [INSERT REFERENCE TO HARRIETS SECTION ON SUB NEPTUNE COMPOSITION CALCULATION]. From these mixing ratios, the mean molecular weight is calculated as $\mu = \sum_i X_i \, m_i$.

The absorption cross sections ($\sigma_{\lambda,i}(P,T)$) are loaded as arrays and we use RegularGridInterpolator to evaluate the cross sections at the pressures and temperatures of the model atmosphere and at the wavelengths of interest. For each wavelength λ_k and layer j, the total absorption per unit path length is then calulated in equation 8.

$$\sum_i n_{i,j} \, \sigma_{\lambda_k,i}(P_j, T_j) = n_j \sum_i X_i \, \sigma_{\lambda_k,i}(P_j, T_j) \tag{8}$$

1.4 Collision-Induced Absorption

The script also accounts for collision-induced absorption (CIA) from H_2–H_2 and He–H_2 pairs. H_2 has no permanent dipole moment, so dipole transitions are very weak. However, during a collision between molecules the electron clouds of the pair are briefly distorted, creating an induced dipole that can form a momentary bond. This is known as london dispersion forces and is a key mechanism for absorption in H_2-rich atmospheres. This short-lived collision pair can absorb radiation in the wavelength we are observing.

This phenomenon is important in H_2-rich atmospheres, where the line opacity from H_2 alone is weak but the CIA contribution can become significant as it scales with the square of the number density. Unlike molecular absorption lines, which are narrow and defined, collision-induced absorption produces a broad and smooth absorption band as collisions have a range of random separations and interaction geometries rather than a single bound-state transition like a typical molecular line. As a result, collision-induced absorption often fills in spectral windows between strong molecular bands which makes detecting these molecules more difficult. This is important for our project as we are looking for ideal sub-neptune tagets to study the atmospheres of.

Because CIA is a binary process, its contribution to τ_λ is proportional to the collision rate and therefore scales with the square of the number density for H_2–H_2 and the product of the number densities for He–H_2. For a background gas with total number density n_j and mixing ratios X_i, this gives a quadratic density dependence ($\propto n_j^2$), not the linear scaling used for one-body line absorption:

$$\tau_\lambda^{\mathrm{CIA}} \propto n_j^2\, X_{\mathrm{H}_2}^2\, \sigma_\lambda^{\mathrm{H}_2-\mathrm{H}_2} + n_j^2\, X_{\mathrm{H}_2} X_{\mathrm{He}}\, \sigma_\lambda^{\mathrm{He}-\mathrm{H}_2} \tag{9}$$

Since $n_j = P_j/(k_B T_j)$, CIA generally becomes more important at higher pressure (deeper layers), where collisions are more frequent. After these CIA terms are added to the line-by-line molecular opacity, the code holds a 2D array opacity[k,j] representing $\sum_i n_{i,j}\sigma_{\lambda_k,i}$ for each wavelength and pressure level.

Due to CIA having a low probability of absorption per collision, it is often negligible in low abundance gases, which is why we only include it for H_2–H_2 and He–H_2 pairs in our model, as these are the dominant molecules in a sub-neptune atmosphere and therefore have the highest collision rates.

1.5 Slant Optical Depth and Transit Integration

Given the radial grid r_j and the opacity at each layer, the next part of our script finds the optical depth $\tau_\lambda(r)$ along rays that pass through the planet at different layers r. For a ray with minimum radius r_i, the path length through layers at larger radii is set by equation 10 so that the path length across a shell between r_j and r_{j+1} is $\Delta s_j = s(r_{j+1}) - s(r_j)$.

$$s(r) = \sqrt{r^2 - r_i^2} \tag{10}$$

The code loops over the layers above each impact parameter and approximates the slant optical depth by the sum 11 where the opacity is averaged between adjacent layers to obtain an estimate.

$$\tau_\lambda(r_i) \approx \sum_j \left[\frac{\mathrm{opacity}_{k,j} + \mathrm{opacity}_{k,j+1}}{2}\right] \Delta s_j \tag{11}$$

If the pressure at a given radius exceeds the prescribed cloud-top pressure P_{cloud}, a very large additional optical depth (e.g. $\tau \sim 10^3$) is added to mimic an effectively opaque cloud deck.

The fraction of light transmitted along that ray is $e^{-\tau_\lambda(r_i)}$. The script converts this to an effective transit radius with the standard annulus-integration formula for transmission spectroscopy in equation 12.

$$\Delta_\lambda = \frac{\pi R_{\mathrm{p}}^2 + 2\int_{r>R_{\mathrm{p}}} r\left[1 - e^{-\tau_\lambda(r)}\right] \mathrm{d}r - 2\int_{r<R_{\mathrm{p}}} r\, e^{-\tau_\lambda(r)}\, \mathrm{d}r}{\pi R_{\mathrm{s}}^2}. \tag{12}$$

The first term represents the area of an opaque disk of radius R_{p}. The second term adds on the contribution from atmospheric annuli above this radius, weighted by the fraction of light they block. The third term subtracts light that is able to leak through layers that are geometrically inside R_{p} but not completely opaque because of the finite optical depth. In practice these integrals are again evaluated numerically using the discrete r_i grid and the precomputed values of $e^{-\tau_\lambda(r_i)}$.

The result is a model transit depth Δ_λ as a function of wavelength, which the notebook plots in parts-per-million (ppm). Changing the planetary radius, mass, temperature, mean molecular weight, or cloud-top pressure directly affects the computed scale height and optical depths, and therefore changes the amplitude and shape of the spectral features.

References

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